ICE - Algo, Assignment.

The goal of this assignment is to price an option using MonteCarlo simulation.

The underlying has value S_t at time t.

We consider the following payoffs :

- call option : $p(S) = Max(S - K, 0)$, where K is the strike of the option
- put option : $p(S) = Max(K - S, 0)$, where K is the strike of the option
- straddle : $p(S) = |S - K|$, where K is the strike of the option
- square : $p(S) = S^2$

Please educate yourself on what an option is and what a payoff is in this context, if not already familiar.

We consider 2 models used in pricing such options:

1/ LogNormal model :

$$S_T = S_0 \exp\left(-\frac{\sigma^2 T}{2} + \sigma X\sqrt{T}\right)$$

2/Normal model

$$S_T = S_0 + \sigma X\sqrt{T}$$

For both models $X \sim \mathcal{N}(0,1)$ is a standard normal Gaussian random variable. This means that in the MonteCarlo simulation explained below the random numbers used will have an expectation (average) of zero, and will be normally distributed around zero with one standard deviation being equal to 1.

Please lookup any terms you are not familiar with, or else send me a question.

The price today of an option struck at K and maturing at T can be computed as :

$$price = E[p(S_T, K)]$$

This price can be computed through MonteCarlo simulation :

For N simulations :

Generate N standard normal Gaussian random variable : X_1, X_2 …

For each simulation compute $S_{T,1}, S_{T,2}$ … (depending on the model)

Compute $p_i = p(S_{T,i}, K)$ according to the payoff type, as specified above

Compute the price as the average of the $p_i's$:

$$price = \frac{1}{N}\sum_{i=1}^{N} p_i$$

Write a program that takes as input :

- S_0 (the value of the underlying today)
- T (the time to maturity)
- K (the strike)
- σ (the volatility)
- $OptionType$: Call, Put or Straddle, Square
- $ModelType$: LogNormal or Normal
- N the number of simulation in the MonteCarlo algorithm

The program should return the price.

The program should use OOP to handle OptionTypes and ModelTypes.